UNIVERSITY OF WESTERN ATTICA

DEPARTMENT OF ENGINEERING INFORMATION AND COMPUTER

DIGITAL DESIGN

WORKSHOP 4
REGISTER-SLIDES

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CONTENTS

CHAPTER 1

A FEW WORDS ABOUT WORK

The work relies on sliders and registers. More specifically, in the four-bit register with D flip - flop , right shift register, left shift register and general purpose shift register (Integrated 74 LS 194). Their operation is presented through the " multisim " simulator, specifically the " SIPO " shift register operation with serial input and parallel output, the " PISO " shift register operation with parallel input and serial output, the general purpose shifter operation and finally the operation of the eight-bit register and transfer of information from four-bit to four-bit register.

CHAPTER 2

BIBLIOGRAPHY

- Design and implementation of logic circuits
  P. Giannakopoulos L. Aslanoglou

CHAPTER 3

DESCRIPTION OF WORK IMPLEMENTATION

To implement the work, the " multisim " simulator was used to design the circuits. In more detail, cables, grounds, VCC sources , lamps, logic gates (AND , OR , NOR , NAND , XOR , XNOR , NOT), oscilloscope, the integrated 74 LS 194 N and the " CLEAR " and " CLOCK " switches were used.

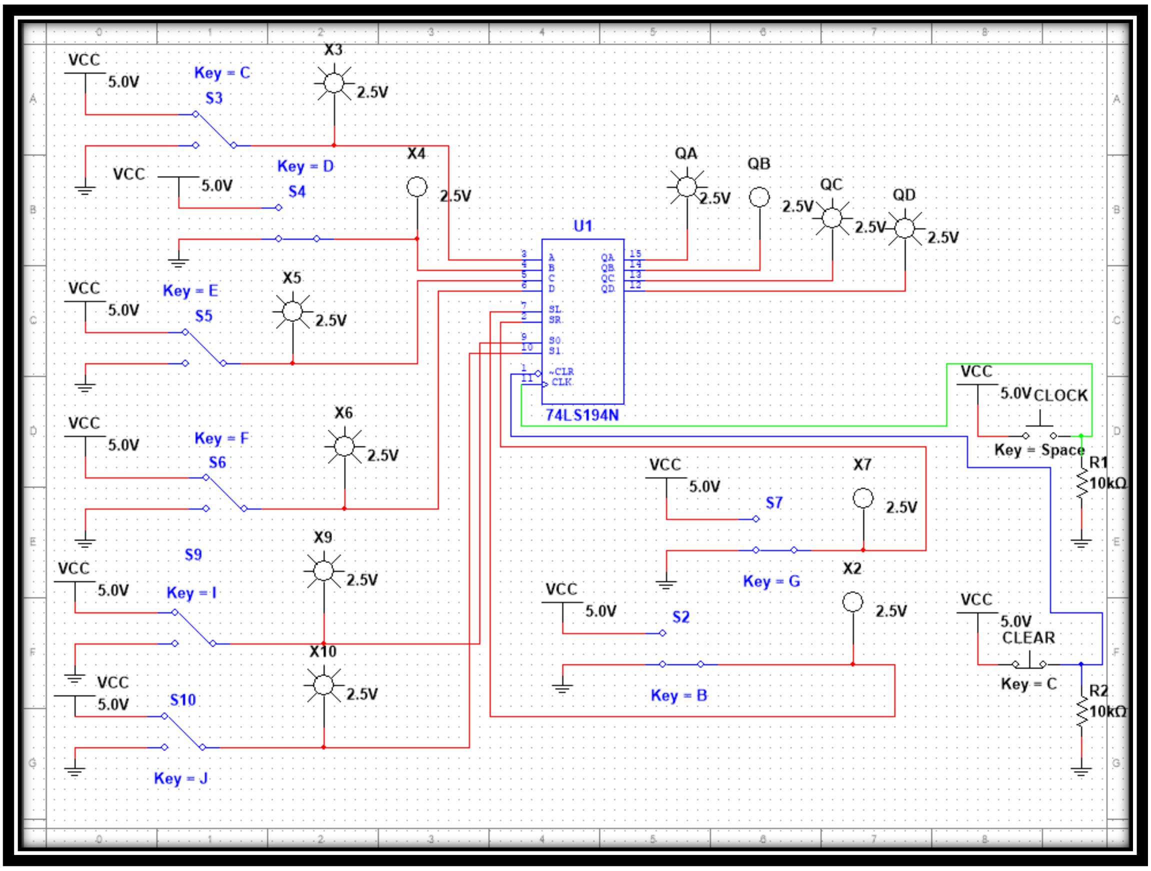
CHAPTER 4

EXERCISES

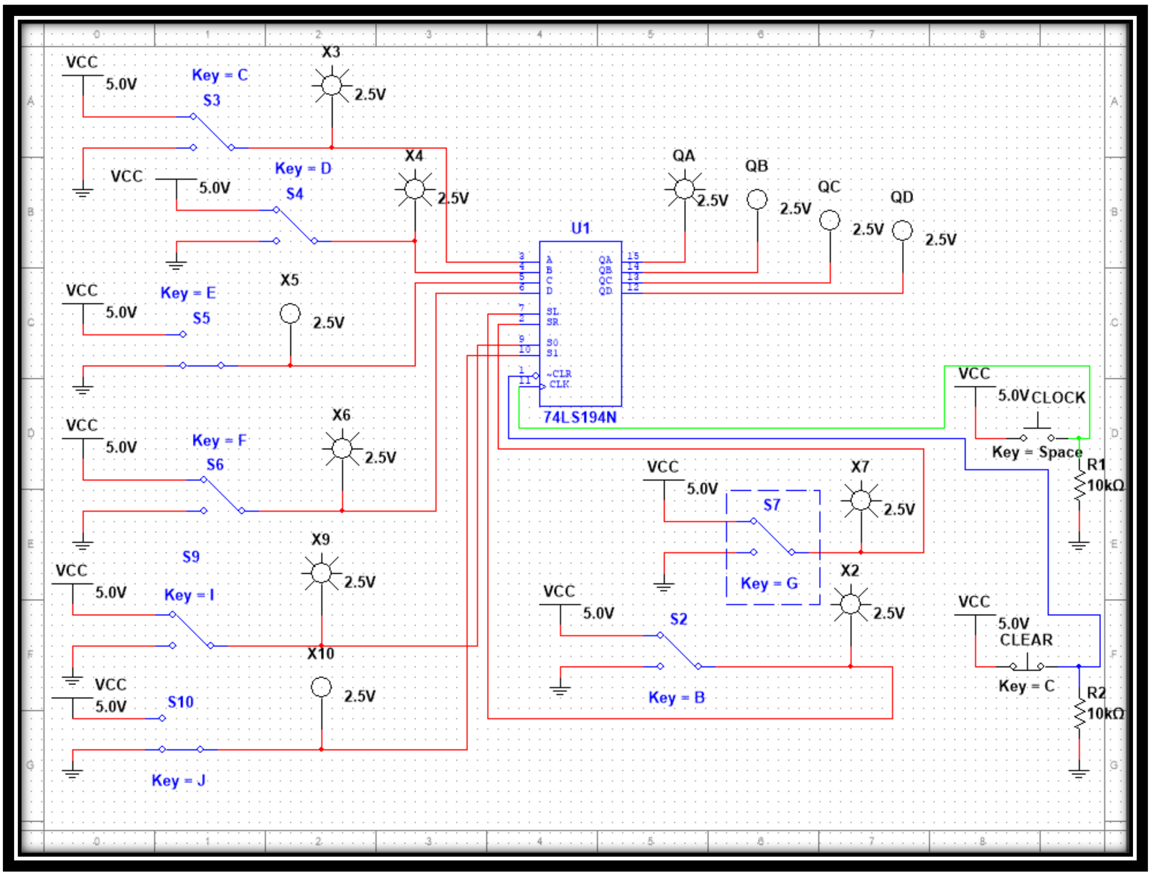
EXERCISE 6.2.2 GENERAL PURPOSE SLIP REGISTER 74194

THEORETICAL RESULTS

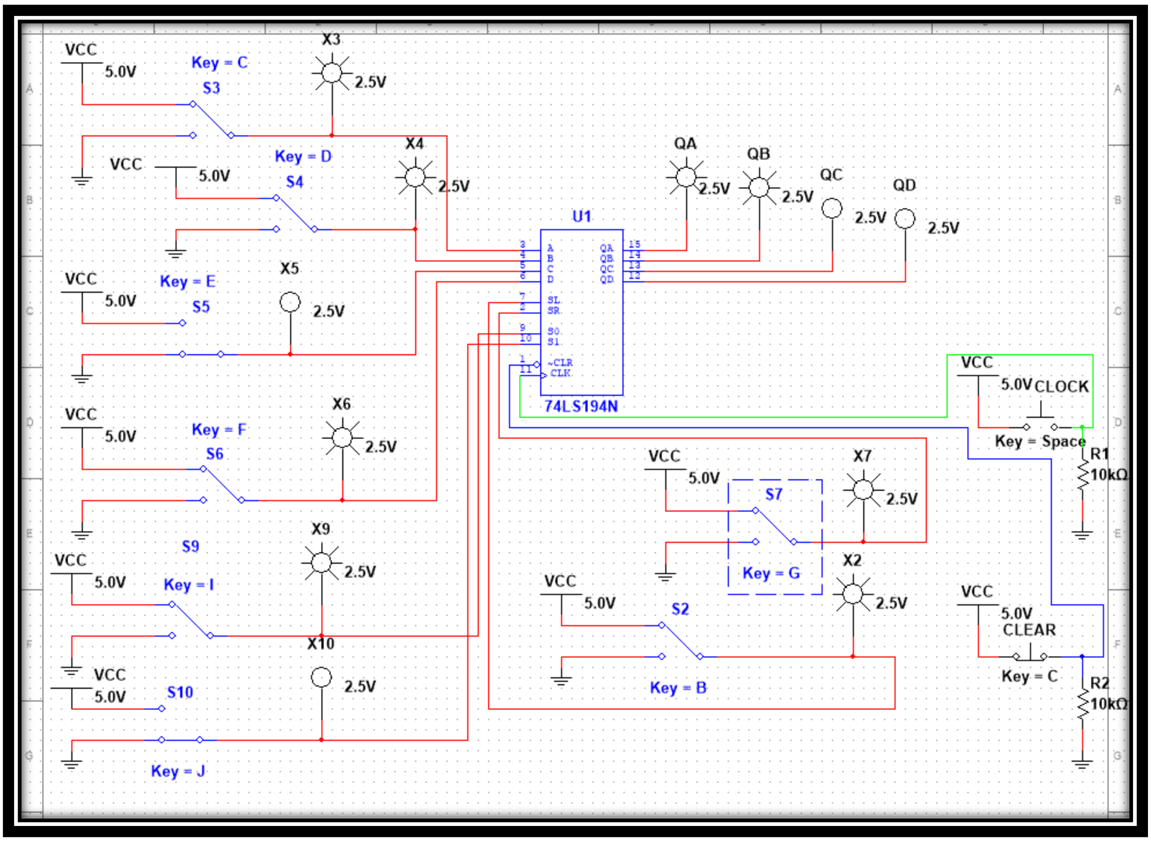
1)



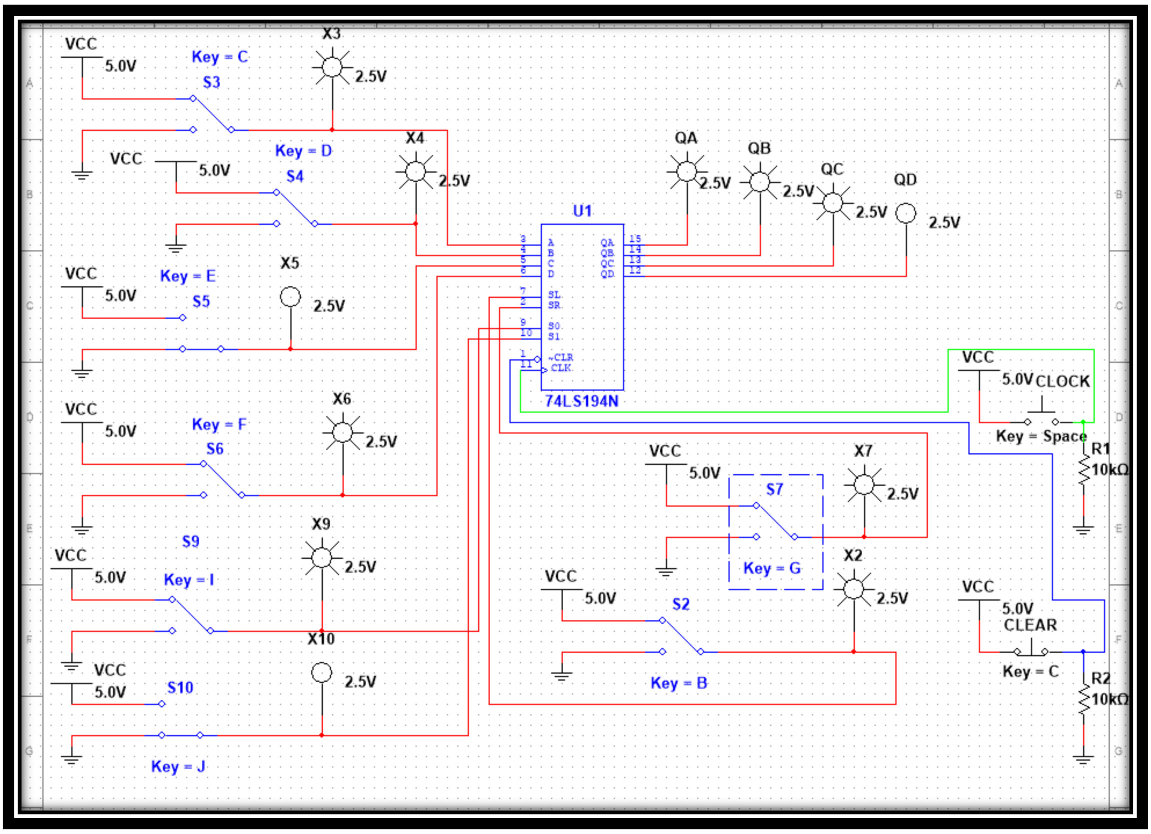
2) After the first clock pulse



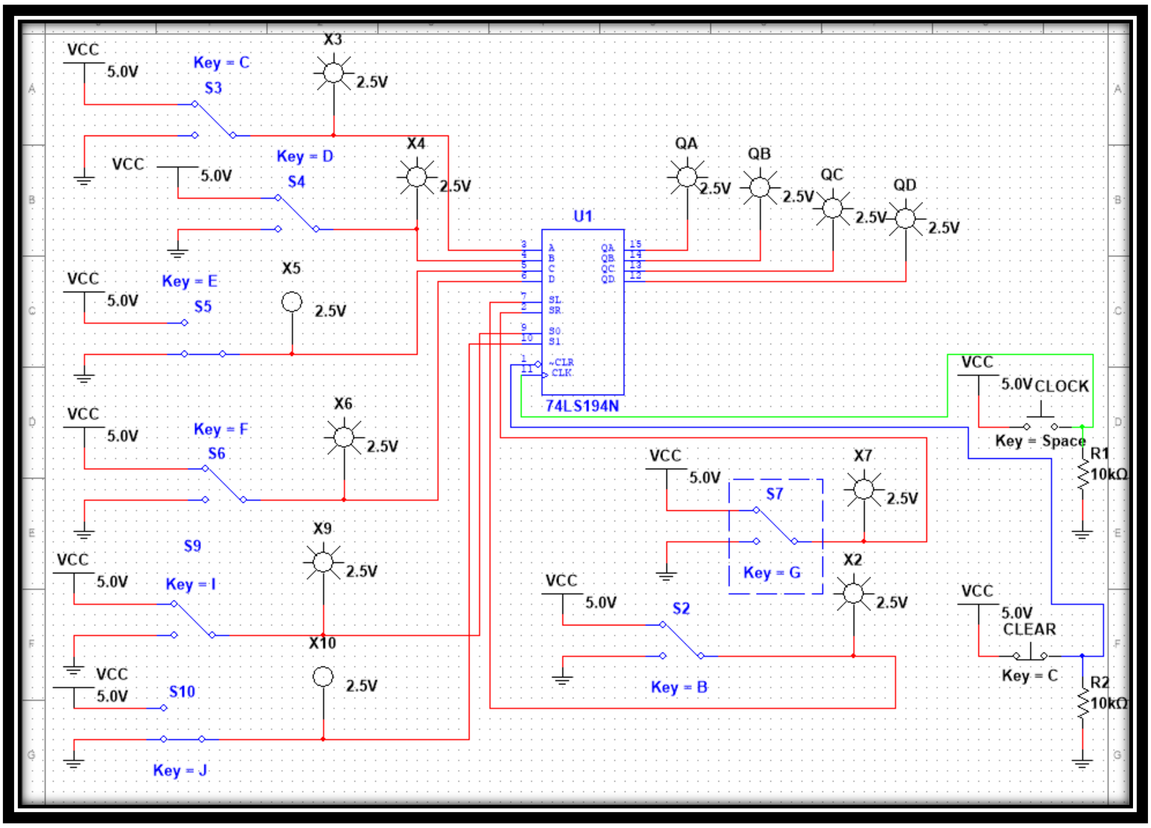
After the second clock pulse



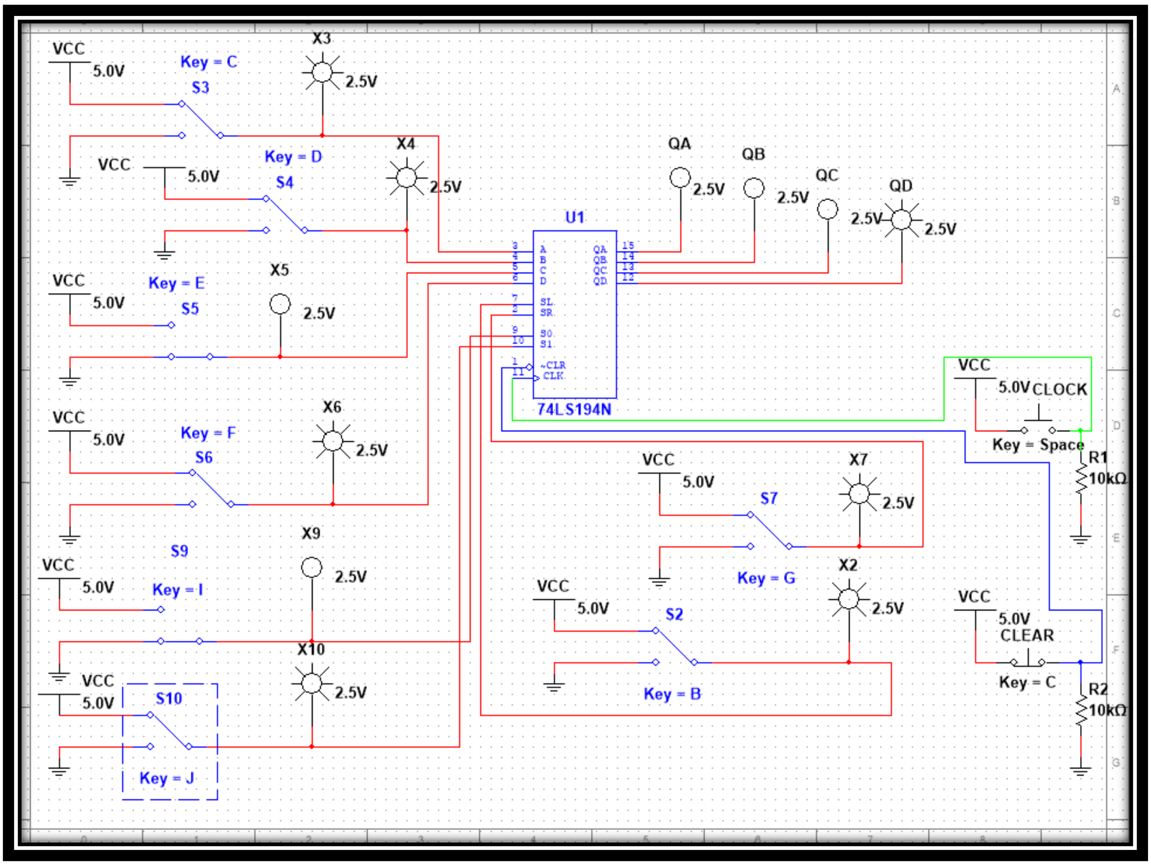
After the third clock pulse



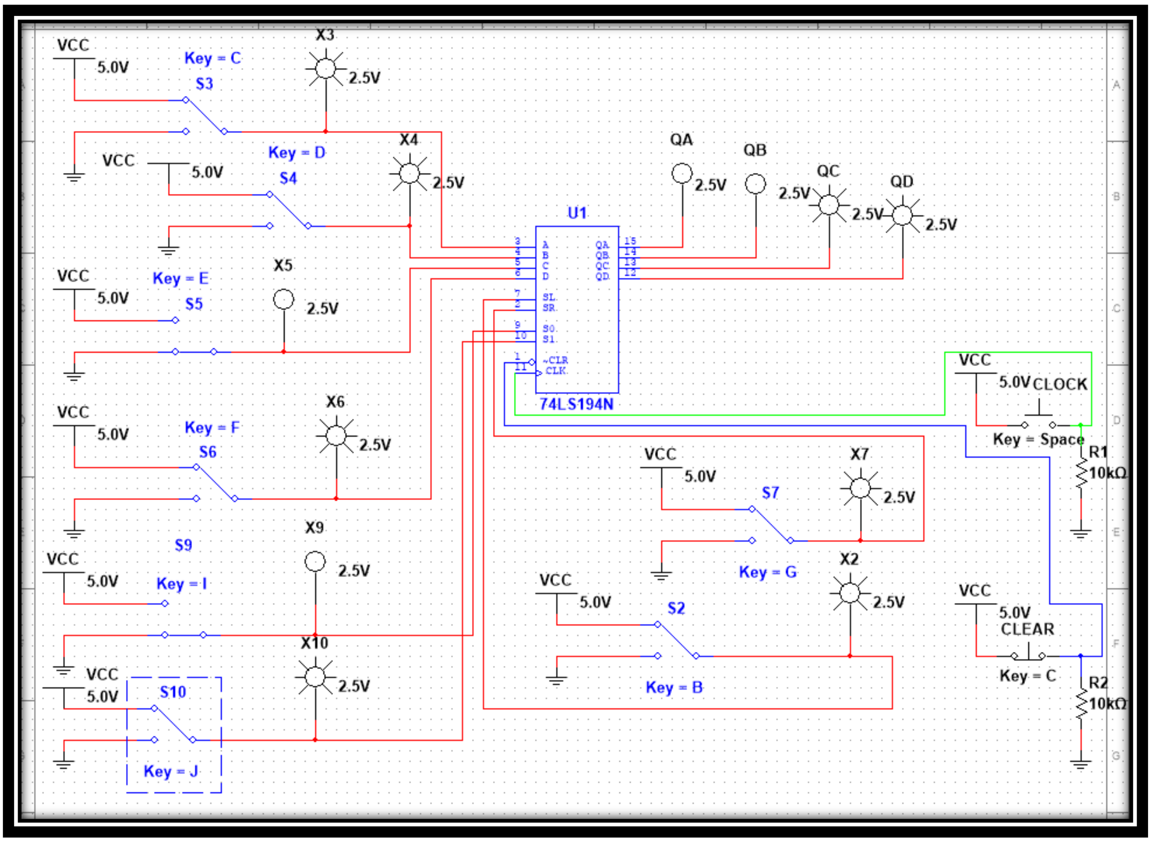
After the fourth clock pulse



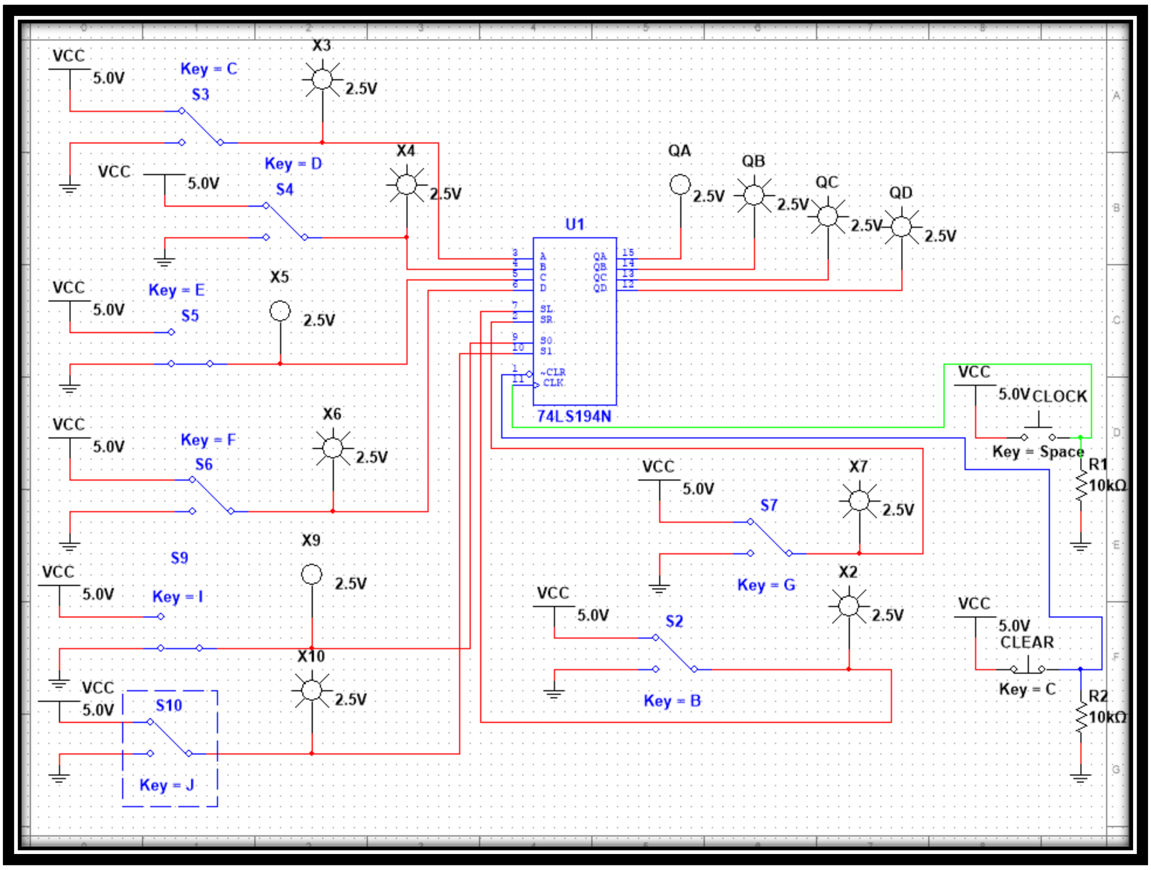
3) After the first clock pulse



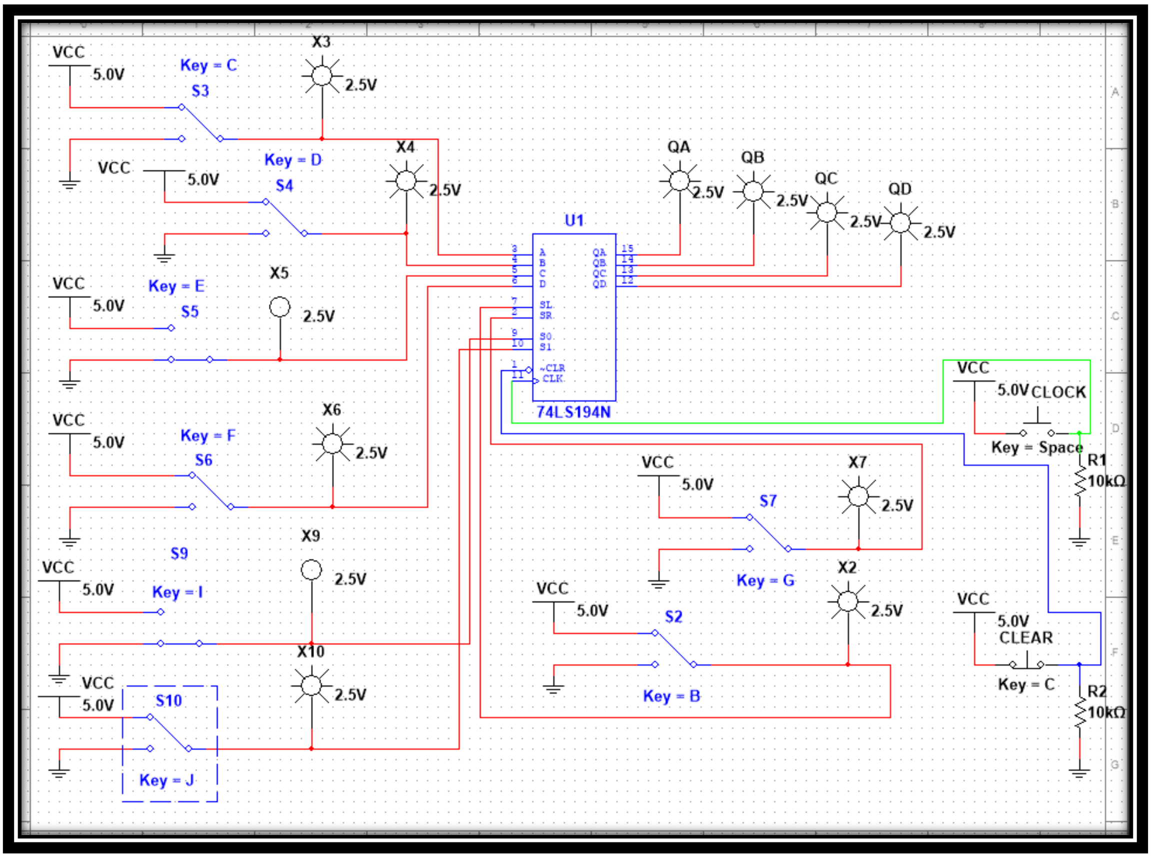
After the second clock pulse



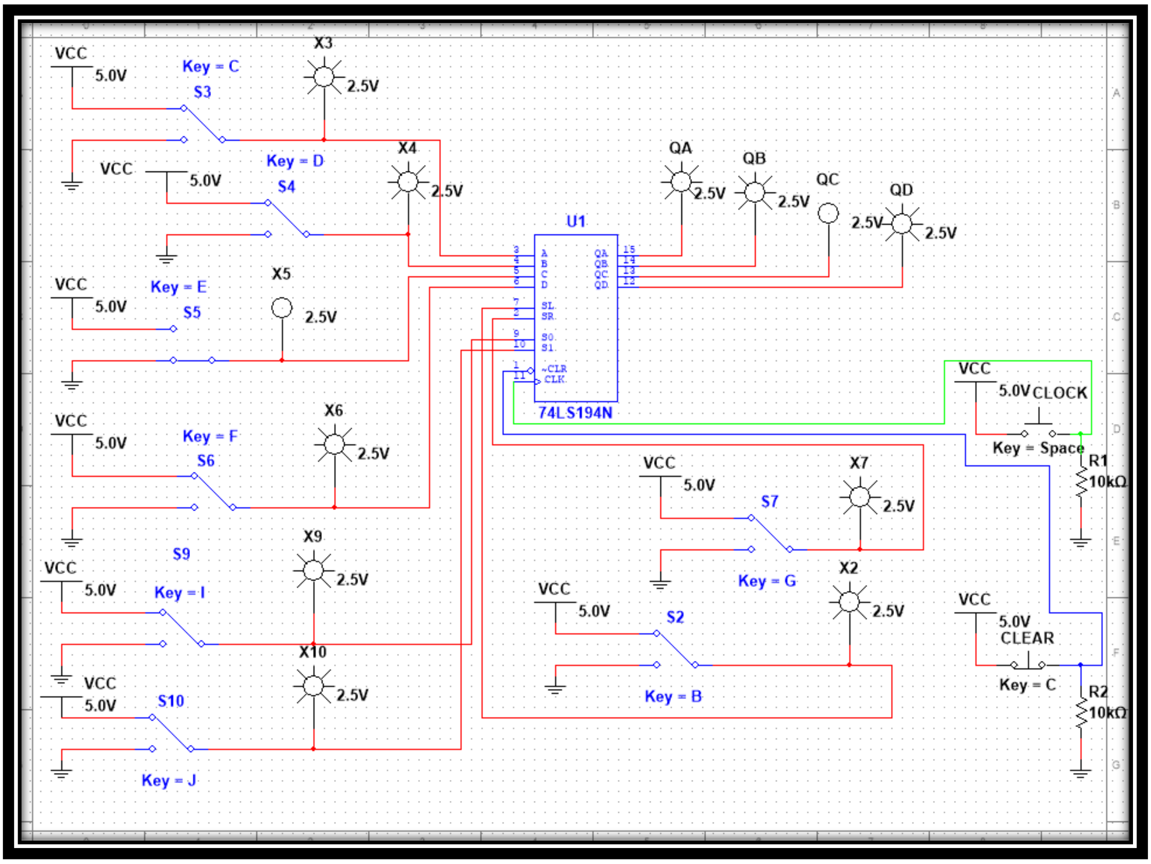
After the third clock pulse



After the fourth clock pulse

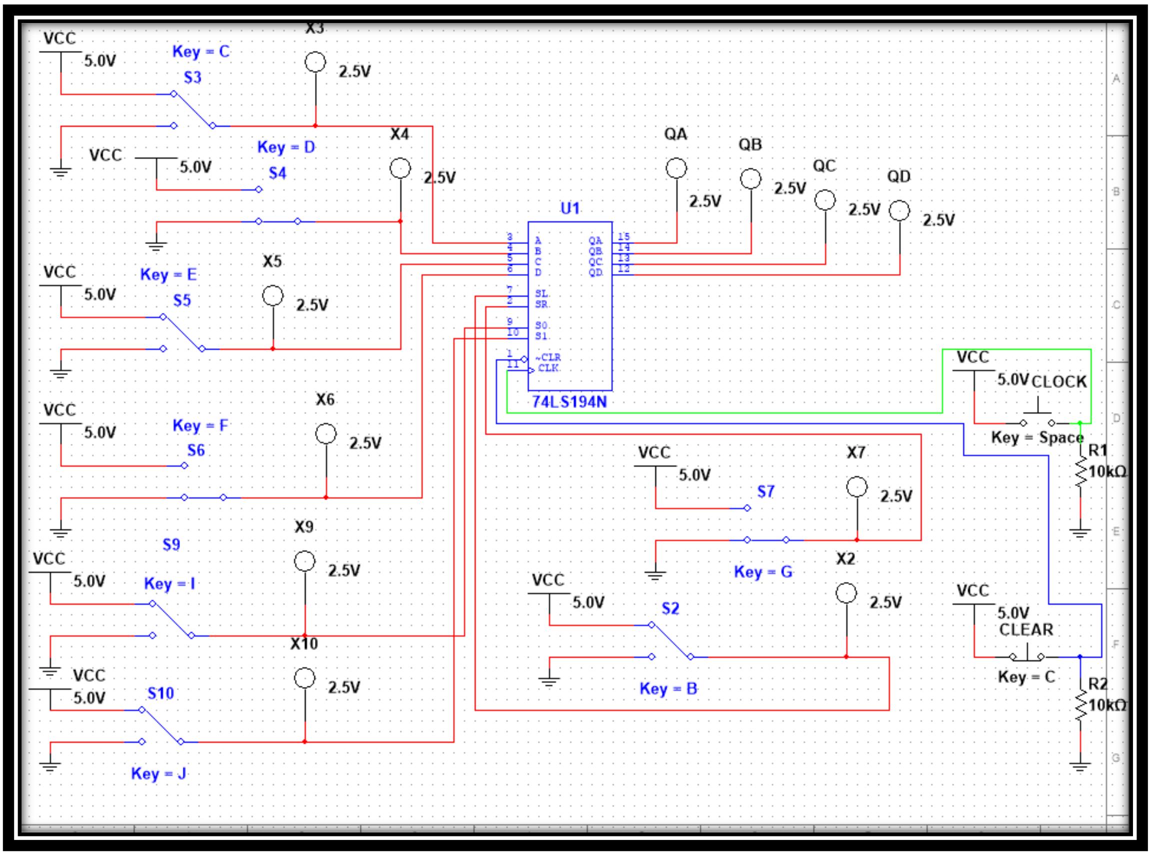


4)



With the inputs " S 1" and " S 0" at logical "1", giving a " clock " pulse we observe a 4 Bit parallel input . In other words, the led lamps of the outputs get the corresponding power corresponding to the led lamps of the inputs, for example the lamp of the input " A " is open and therefore the corresponding one of the output "QA".

SIMULATION

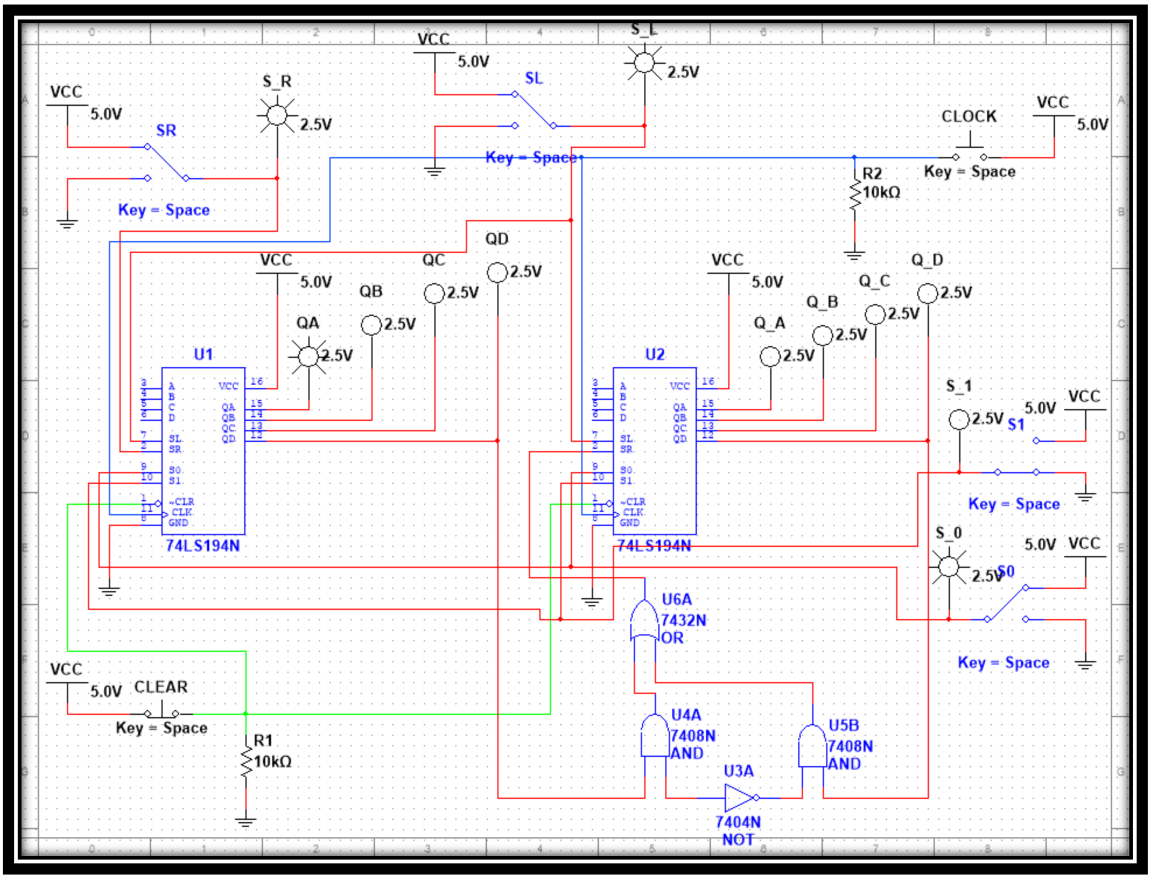


VCC
5.0V
Key = C
S3
X3
2.5V
Key = D
S4
X4
U1
74LS194N
QA
QB
QC
QD
Key = E
S5
X5
Key = F
S6
X6
S9
Key = I
X9
X10
S10
Key = J
S7
X7
Key = G
X2
S2
Key = B
CLOCK
Key = Space
R1
10kΩ
CLEAR
R2

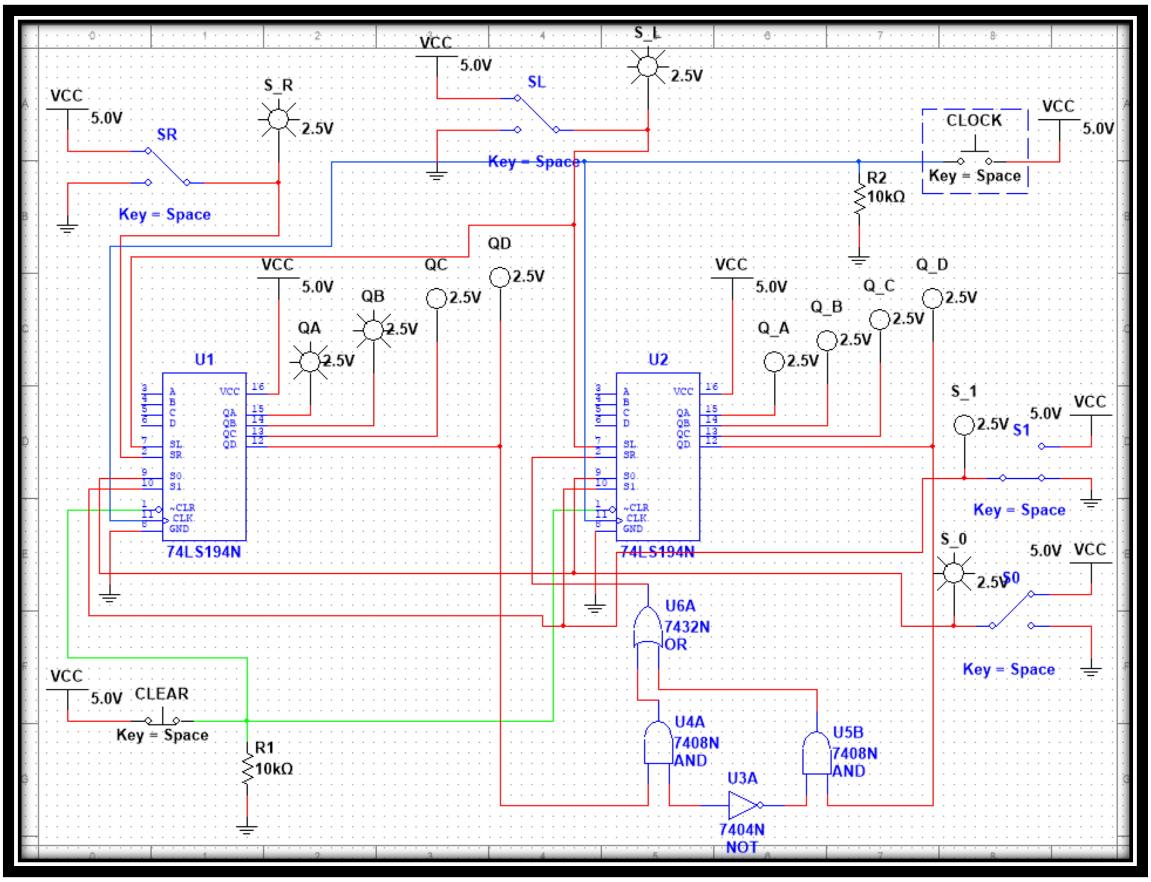
EXERCISE 6.2.3 OCTABIT REGISTER AND TRANSFER OF INFORMATION FROM QUAD BIT TO QUAD BIT REGISTER

THEORETICAL RESULTS

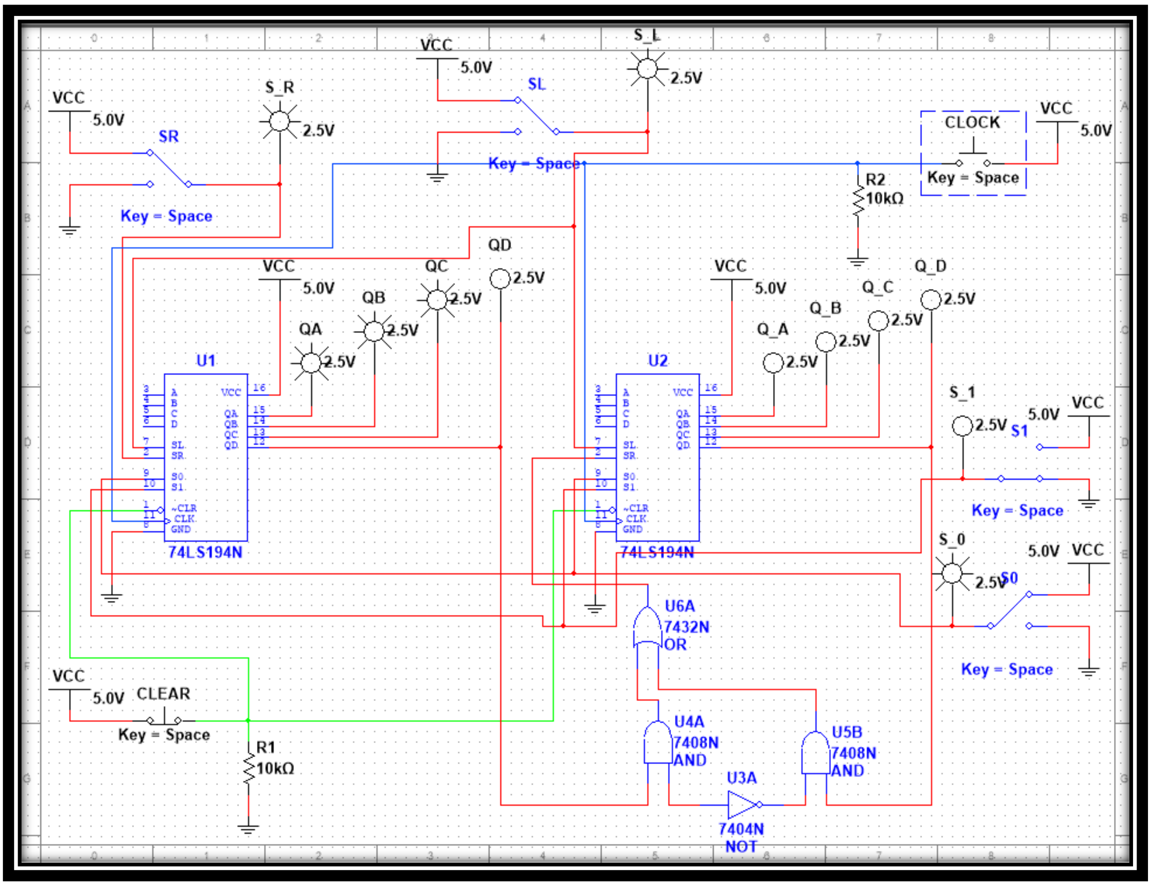
1) After the first clock pulse



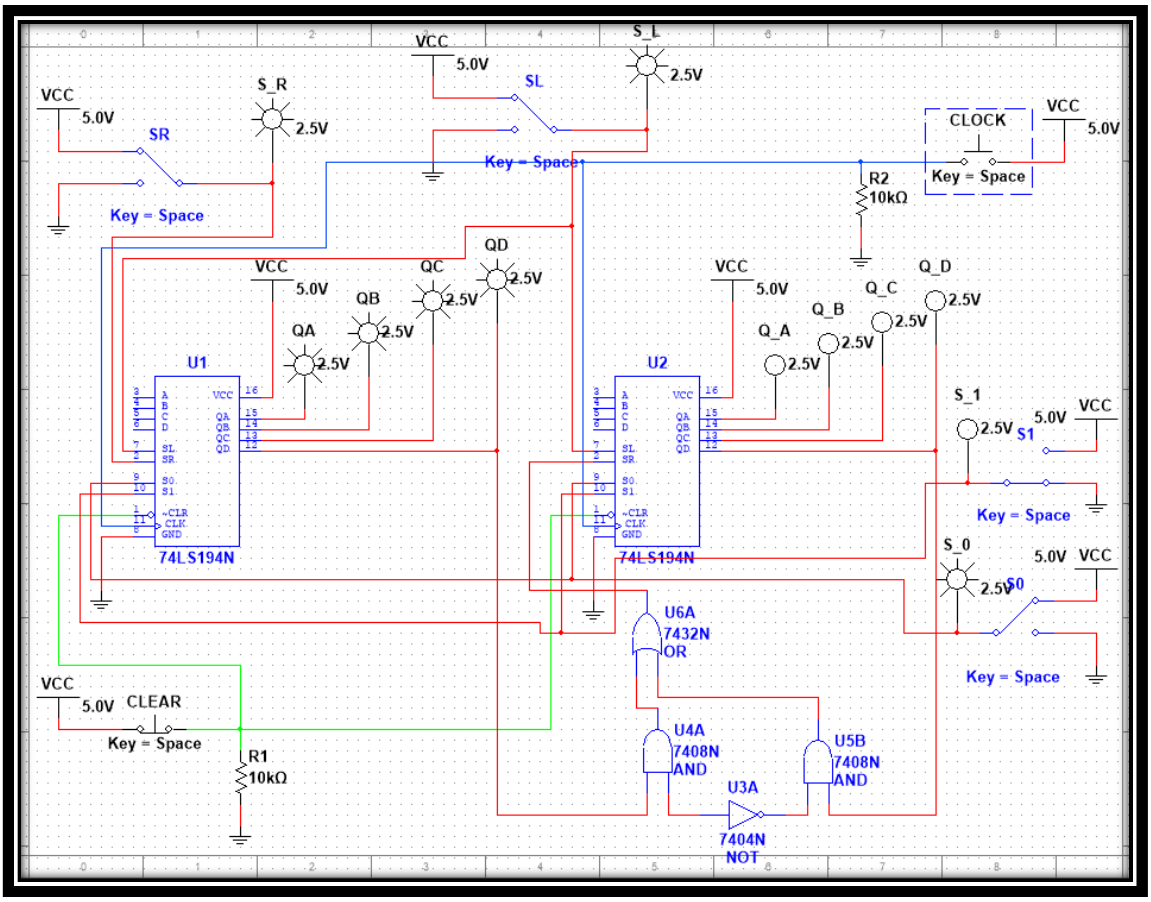
After the second clock pulse



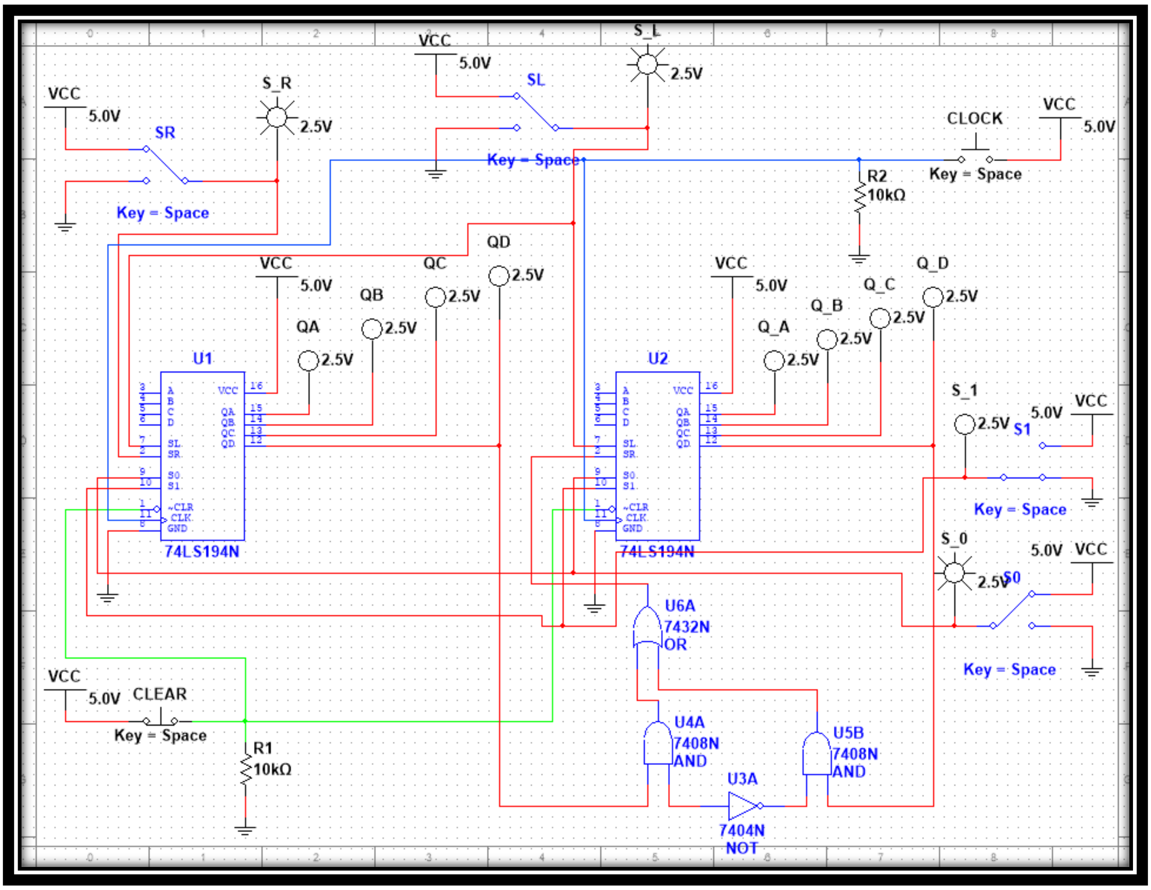
After the third clock pulse



After the fourth clock pulse


VCC
5.0V
SR
S_R
2.5V
Key = Space
SL
S_L
CLOCK
Key = Space
R2
10kΩ
QA
QB
QC
QD
U1
74LS194N
U2
74LS194N
Q_A
Q_B
Q_C
Q_D
S_1
S1
S_0
S0
U6A
7432N
OR
U4A
7408N
AND
U5B
7408N
AND
U3A
7404N
NOT
CLEAR
R1
10kΩ

SIMULATION



VCC
5.0V
S_L
2.5V
SL
S_R
2.5V
SR
Key = Space
CLOCK
R2
10kΩ
QD
QC
QB
QA
U1
74LS194N
U2
74LS194N
Q_D
Q_C
Q_B
Q_A
S_1
S1
S_0
S0
U6A
7432N
OR
U4A
7408N
AND
U5B
7408N
AND
U3A
7404N
NOT
CLEAR
R1
10kΩ